Nithin Gopalakrishnan Nair

iamnithingk@gmail.com | +1-667-212-9785 | Website | Scholar | LinkedIn | GitHub

Research Focus

Large-scale training of generative and multimodal models, with a focus on synthetic data, compositional generation, and 3D understanding.

Research Experience

Apple – Research Engineer, Multimodal Agents Jun 2025 – Present

- Improved multimodal model performance on **human interaction understanding** tasks.
- Own **agent development infrastructure**, integrating simulation and evaluation workflows.
- Define evaluation protocols and scoring rubrics to assess multimodal agent behavior and guide model improvements.

Google – Google Beam / Project Starline Dec 2024 – Mar 2025

- Demonstrated the first successful training of transformer-based novel view synthesis models with **diffusion-generated synthetic data**.
- Introduced source/target token disentanglement to suppress synthetic-data artifacts and improve generalization to unseen scenes (ICCV 2025; patent application filed).

NVIDIA Research Apr 2024 – Dec 2024

- Developed the first foundation-model framework for **physical-property assignment to 3D meshes** through part segmentation and multimodal material identification.
- Co-inventor on U.S. Patent Application 19/024,394, filed January 2025.

Adobe Inc. May 2023 – Aug 2023

- Built a training-free diffusion pipeline for interactive image editing.

Mitsubishi Electric Research Labs (MERL) May 2022 – Aug 2022

- Developed energy-based guidance for zero-shot conditional generation with pretrained diffusion models (Steered Diffusion, ICCV 2023).

Research Highlights

Inference-Time Steering and Compositional Generation 2022 – 2024

- Introduced **Steered Diffusion**, the first plug-and-play framework to unify image-to-image and label-conditioned generation through **inference-time steering**, without task-specific training (ICCV 2023).
- Pioneered **compositional multimodal generation** with **Unite and Conquer**, combining independently trained diffusion models at inference time without joint multimodal training (CVPR 2023).
- Extended pretrained text-to-image models to multiple simultaneous conditions through **variance-guided feature fusion** (**MaxFusion**, ECCV 2024).

Efficient Diffusion Steering (D3SG) 2026

- Developed **D3SG** for guidance without U-Net backpropagation, combining noise-space and image-space gradients with adaptive scaling for linear and nonlinear conditions. *Research manuscript.*

Synthetic Data for Generalizable Image Restoration (GenDeg) 2025

- Introduced **GenDeg** to synthesize realistic image degradations from clean inputs, improving **all-in-one restoration on unseen degradation patterns** (CVPR 2025).
- Built **GenDS (750K+ samples)**, including **550K+ synthetic image pairs** spanning six degradation types.

IARPA BRIAR – Unconstrained Face Recognition 2021 – 2025

- Achieved **state-of-the-art results on five face-recognition benchmarks** through large-scale pretraining, staged fine-tuning, and context-aware face crops (FG 2025; co-first author).
- Surpassed AdaFace by **31.4 percentage points in BRIAR TAR at FAR** 10^{-3}, with both models pretrained on WebFace12M; achieved **81.38% Rank-1** with WebFace42M.

Selected Publications

18 peer-reviewed publications; full list on Google Scholar.

1. **Scaling Transformer-Based Novel View Synthesis with Token Disentanglement and Synthetic Data** 2025
   Nithin Gopalakrishnan Nair*, Srinivas Kaza*, Xuan Luo, Vishal M. Patel, Stephen Lombardi, Jungyeon Park
   IEEE/CVF International Conference on Computer Vision (ICCV)

2. **GenDeg: Diffusion-Based Degradation Synthesis for Generalizable All-in-One Image Restoration** 2025
   Sudarshan Rajagopalan, **Nithin Gopalakrishnan Nair**, Jay N. Paranjape, Vishal M. Patel
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

3. **MaxFusion: Plug&Play Multi-Modal Generation in Text-to-Image Diffusion Models** 2024
   Nithin Gopalakrishnan Nair, Jeya Maria Jose Valanarasu, Vishal M. Patel
   European Conference on Computer Vision (ECCV)

4. **CrowdDiff: Multi-hypothesis Crowd Density Estimation using Diffusion Models** 2024
   Yasiru Ranasinghe, **Nithin Gopalakrishnan Nair**, Wele Gedara Chaminda Bandara, Vishal M. Patel
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

5. **Steered Diffusion: A Generalized Framework for Plug-and-Play Conditional Image Synthesis** 2023
   Nithin Gopalakrishnan Nair, Anoop Cherian, Suhas Lohit, Ye Wang, Toshiaki Koike-Akino, Vishal M. Patel, Tim K. Marks
   IEEE/CVF International Conference on Computer Vision (ICCV)

6. **Unite and Conquer: Plug&Play Multi-Modal Synthesis Using Diffusion Models** 2023
   Nithin Gopalakrishnan Nair, Wele Gedara Chaminda Bandara, Vishal M. Patel
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

7. **PETALface: Parameter Efficient Transfer Learning for Low-Resolution Face Recognition** 2025
   Kartik Narayan, **Nithin Gopalakrishnan Nair**, Jennifer Xu, Rama Chellappa, Vishal M. Patel
   IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), ***Oral***

* Equal contribution.

Education

Johns Hopkins University, Baltimore, USA 2021 – 2025

- Ph.D. in Electrical and Computer Engineering, advised by Prof. Vishal M. Patel; GPA: *3.96/4.0*.

Indian Institute of Technology Madras, Chennai, India 2015 – 2020

- B.Tech. and M.Tech. Dual Degree in Electrical Engineering; CGPA: *9.0/10.0*.

Technical Skills

Languages: Python, C, C++, MATLAB
Frameworks: PyTorch, JAX
Generative Models: Diffusion models, autoregressive models, VAEs, GANs

Mentoring

Mentored five student researchers at Johns Hopkins University (2022–2025).

Academic Service

Reviewer: ICLR, NeurIPS, CVPR, ICCV, ECCV; IEEE TPAMI, TIP, and TMI.